

Predicting the Balance of Weapons in a Top-Down Shooter Game

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1 Introduction

In game design, game balance has a significant impact on player experience and engagement. It maintains an optimal difficulty level, ensuring players avoid frustration or boredom and consistently feel a sense of challenge. However, achieving weapon balance is challenging for game designers because it requires a deep understanding of both objective game-play metrics and subjective player perception. If an imbalanced game is created or imbalanced features are created, this can also lead to the use of one single optimal strategy[8], which is, in general, not wanted by designers.

Traditional game balance approaches are based on designer intuition, which is usually followed by iterative testing, but it can be time-consuming and may not fully capture the subtle relationship between weapon statistics, player performance, and player satisfaction. Therefore, predicting how adjusting weapon attribute parameters affects player perception and weapon preference is difficult.

This paper proposes a data-driven weapon balance evaluation method that integrates objective weapon attributes, in-game player performance, and subjective player feedback. We first developed a 2D shooter game called "TopDownShooter", which features four different weapons, including rifle, sniper rifle, shotgun, and SMG, with distinct attributes. Then we collected quantitative performance data through a controlled gameplay experiment, and qualitative feedback via post-game questionnaires to measure player satisfaction with the weapon.

Our primary contribution is a machine learning framework that predicts weapon balance based on weapon statistics. We create a supervised learning dataset by labeling weapons using a formula that combines objective player performance metrics and subjective player feedback. The trained model predicts whether players would consider a weapon with given attributes to be balanced and desirable, enabling designers to evaluate potential weapon configurations before conducting extensive playtesting.

Section 2 reviews related work in game balance and player modeling. Section 3 introduces the design of TopDownShooter. In sections 4 and 5, the game AI approach and its implementation are explained. Then, section 6 evaluates our approach through experimental results. Finally, in sections 7 and 8, the advantages and limitations of our approach are discussed and concluded.

2 Background and Related Work

Many papers have attempted to reconcile the topic of game balance as a measure and game balancing as a process for video games. Becker et al. [2] provide a literature review that illuminates the various interpretations of balance, claiming that while there is no

strictly agreed-upon definition for balancing, most authors agree upon the key features of a well-balancing game, those being the perceived difficulty, player skill, and meaningful decision options being provided to players. The paper also explores the interaction between individual game elements influencing the overall balance of an experience, providing further basis for our approach. Meanwhile, flow theory [3] further explains the relationship between player experience and balance, thus emphasizing the importance of maintaining balance.

For this paper, we adopt the definition of game balancing proposed by Volz et al. [11], who describe it as "the modification of parameters of the constitutive and operational rules of a game [...] in order to achieve optimal configurations in terms of a set of goals." In our research, the primary goal of balancing is to fulfill players' needs for satisfaction, appropriate difficulty, meaningful decision-making, and challenge. The papers cited following this may use differing methods and approaches to suit their needs, but ultimately provide a useful palette of methods that game developers can adapt to their own contexts.

Player modeling has been used in various methods and processes for games in various ways in the past. Andrade et al. [1] applied the player modeling approach to adjust difficulty dynamically by training Reinforcement Learning agents to emulate human players and modify their behavior in an experimental fight-em-up game. Although our approach is not based on dynamic adjustment of difficulty, the approach used by Andrade et al. is an illustration of the usefulness of AI approaches of player modeling for the adjustment of difficulty, which is one of the major features of game balance that designers can control.

Pfau et al. [9] approached game balancing ideas in specifically online Player-vs-Player games to explore the issues regarding balance patches and community perception. The paper conducted a case study of players of various levels of skill in the game *Guild Wars 2* and measured each player's opinion as a metric of game balance for deeper insight into community reactions to gameplay changes. Their approach to balance as a data-driven concept is useful for data collection relating to the balance of individual elements of a game.

Karavolos et al. [6] utilized deep learning methods to learn the game balance of in-game levels and the parameters of each player's weapon, analyzing whether there is an imbalance between the two teams of players. Their approach applies the concepts of game balance to entire scenarios in a multiplayer setting to achieve a similar result expected in our method. Their findings also set a precedent for our work, providing accurate and relevant observations. However, the difference in scenarios and applications means that their

approach cannot be used wholesale in our experiment, where the significance of weapon features is much higher within its context.

Since our research is based on weapons balance and, eventually, development around balancing features, the paper by Gravina et al. [5] was of great reference. The authors design a system to procedurally generate custom weapons for the game *Unreal Tournament III*, using weapon features such as rate of fire and efficiency in order to elucidate how balanced a weapon would be during real-time gameplay. Their process is, in essence, a reversal of our method to determine the balance of weapons, using player opinion to hone in on values for weapon properties such as damage and shot spread.

Extending the discussion on weapon balance, Drachen et al. [4] mentioned that the short-range weapons force players to move toward the enemy, thus intensifying combat. However, long-range can be used for a longer distance, which can lead to a less active play style. Besides the rate of fire and efficiency, we should also take this into account.

3 Game

TopDownShooter is a 2D top-down shooter game. The player wins the game if they can kill all enemies without getting too much damage and within the time frame. The player must defeat all enemies in seven rooms. They can move around in those rooms by using the WASD keys, and they can advance to the next room by killing all the enemies in the previous one. They can kill these enemies by shooting bullets with a gun. If one of the enemies gets hit by a bullet, it takes damage, and if the enemy takes enough damage, it will disappear (and is killed).

There are 2 types of enemies: Shooting-type enemies and explosive enemies. Shooting-type enemies have less health; however, they can shoot bullets that will damage the player when they touch them. The enemies that burst upon contact with the player have 2.5 times more health, and they also do 5 times more damage than one bullet. In every room you enter, there will be a different amount and a different combination of enemies; the amount and type of enemies are fixed for every run, and this is not generated. When one of those enemies attacks the player, the player loses health. When the player's health drops to zero or below, or when the time runs out, the player loses. Figure 1 shows an overview of the game.

The player can aim with their gun using the mouse and shoot by clicking or holding the left mouse button. When shooting the enemies, the player can switch between 4 guns using the Q-button:

- Rifle
- Sniper rifle
- Shotgun
- SMG (submachine gun)

Where the sniper rifle is more focused on damage and less on capacity, the SMG emphasizes fire rate and ammo capacity more than damage and range. The shotgun focuses on its core feature, the spread of its bullets when fired. The rifle is the all-around weapon, not focused on a single attribute but averaging across all properties.

The player's interaction with the game will mainly be with these weapons, so the player experience we aim for varies by weapon:

- The rifle is our base, and will function as our independent factor

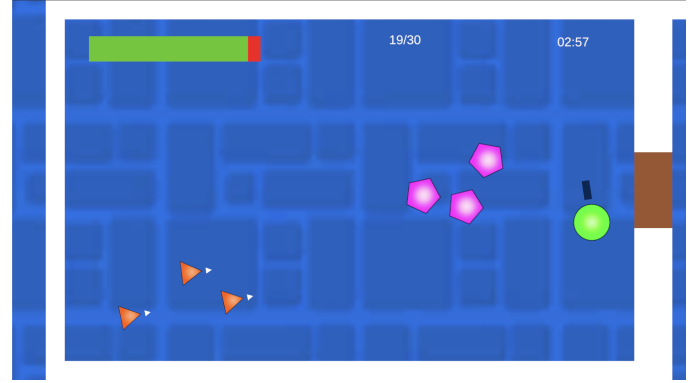


Figure 1: The user interface depicts the player in green, the enemies that explode in pink, and the shooting-type enemies in orange, with a small triangle. The current gun is displayed just below the player as a small black rectangle. The health bar is shown in the upper-left corner, while the timer is shown in the upper-right corner. The door on the right remains closed, as not all enemies have been eliminated.

- The experience for the player when holding the sniper rifle should be the power to kill all enemies with one shot. However, they should also feel the need to be careful how and when they use their ammo
- For the shotgun, we want the player to feel that they are holding a real shotgun, which is also the reason why we focused on the shotgun's core feature.
- For the SMG, the experience is also very focused on the real SMG, where the high fire rate is a characteristic. The player should be able to shoot many bullets. Each bullet deals less damage than the other weapons.

4 Game AI Approach

This project explored how artificial intelligence can help evaluate weapon balance in a 2D player-versus-environment shooting game. The central idea was to predict whether a new weapon design would feel balanced to players. To do this, we modeled the relationship between weapon parameters and players' perceived fairness and enjoyment using supervised learning.

A supervised learning framework was chosen because player feedback naturally provided labeled data. Each weapon test produced questionnaire responses that captured players' sense of fairness, challenge, and enjoyment. From these responses, it became possible to learn how specific weapon attributes shape the overall feeling of balance. Instead of predicting a continuous satisfaction value, we focused on classification. The goal was to explore whether a weapon should be considered balanced or not.

Our theoretical foundation builds on basic models of player experience that link enjoyment to perceived fairness and an appropriate level of challenge. A balanced weapon, in this sense, should neither dominate the game nor feel weak or frustrating to use. After each play session, players completed a short questionnaire of eleven

items. They rated statements about balance, fairness, and satisfaction with weapon behavior on a seven-point scale.

The individual ratings were combined into a Perceived Balance Score (PBS). Rather than averaging the responses, each component received a specific weight reflecting its importance to balance perception. Balance and fairness accounted for about 50 percent of the score, enjoyment and achievement for 20 percent, and satisfaction with weapon attributes such as speed, damage, and range for another 20 percent. The remaining 10 percent came from in-game behavior, represented by how long players used each weapon and how many enemies they defeated with it. These weights were chosen after small pilot tests to make sure the resulting score reflected intuitive differences between well-tuned and unbalanced weapons. Since enjoyment and achievement are inherently subjective aspects of player experience, their inclusion helps the model reflect the psychological factors underlying perceived balance. Previous research has shown that player satisfaction and enjoyment are closely tied to perceived performance and challenge.[7] Similarly, instruments such as the Player Experience Inventory (PXI) demonstrate that players' emotional and cognitive responses are essential indicators of perceived fairness and engagement.[10]

By integrating these insights, the PBS captured not just mechanical fairness but also the emotional and experiential aspects that define a well-balanced weapon from the player's perspective. The PBS ranged from zero to one. Weapons with scores above 0.6 were labeled as balanced (1), and those below 0.4 as unbalanced (0). Samples in between were excluded to reduce ambiguity. This labeling captured how players actually experienced each weapon during gameplay. The input features came from the weapon parameters defined in the game:

- Projectile speed
- Weapon speed
- Weapon damage
- Weapon range
- Number of projectiles
- Weapon spread
- Ammunition capacity
- Reload time
- Critical chance

From these, several derived metrics were created to express higher-level properties such as damage per second, reload efficiency, and overall coverage.

Two standard supervised learning models, logistic regression and random forest, were trained on this dataset. Both learned to estimate the likelihood that a weapon configuration would be perceived as balanced by players. This approach makes it possible to anticipate balance issues early in the design phase, before extensive playtesting, and to use AI as a supportive tool in the iterative process of tuning weapon mechanics.

5 Game AI Implementation

The dataset for this project was collected through a small-scale playtest. 13 players each tested four weapons, resulting in 52 play sessions. Every session included both gameplay logs and a post-play questionnaire. After cleaning the data and removing ambiguous

cases with Perceived Balance Scores between 0.4 and 0.6, 41 valid samples remained for modeling.

Each record combined two types of information. The first part described subjective feedback, including the eleven questionnaire responses that reflected balance, fairness, and enjoyment. The second part contained objective weapon parameters retrieved directly from the game engine. Ratings on the seven-point scale were normalized to values between 0 and 1. This step allowed different players' responses to be compared fairly.

To represent the design properties of each weapon more clearly, several derived features were calculated from the base parameters. Damage and fire rate were multiplied to form damage per second (DPS), a rough measure of offensive power. Ammunition capacity divided by reload time yielded reload efficiency, capturing how quickly a weapon could return to action. The product of projectile speed and range created a coverage variable describing effective reach, while the combination of spread and projectile count represented accuracy trade-offs. These derived variables provided richer information than raw weapon statistics alone.

Behavioral data, such as weapon hold time and number of kills, were also recorded, which are used in the construction of the PBS label. They contributed to the perception of engagement without directly influencing the prediction features.

After preprocessing, all numerical features were standardized using a z-score transformation. The cleaned dataset contained forty-one labeled samples and thirteen numeric predictors. Two models were implemented: logistic regression and random forest. Logistic regression offered a simple baseline for interpreting feature influence, while the random forest provided a non-linear alternative that could capture more complex interactions among weapon parameters.

The model training process followed a three-fold stratified cross-validation procedure. Given the relatively small dataset size, a three-fold split was chosen to balance model stability and data utilization. Each iteration used two-thirds of the data for training and the remaining third for validation, ensuring that every sample served as a validation case once. Evaluation metrics such as accuracy, ROC-AUC, F1-score, and confusion matrices were computed after each fold. The pipeline was designed to be easily reproducible. It could be replicated by following the provided Jupyter notebook, which includes data preprocessing, feature generation, and model evaluation.

This implementation linked subjective player data and objective game metrics into a consistent framework for machine learning.

6 Evaluation

The evaluation examined how well the models could predict whether a weapon would be perceived as balanced by players. Because the dataset contained only 41 valid samples, a three-fold stratified cross-validation method was applied. This ensured that every sample appeared once as a validation case and twice as part of the training data, while preserving the ratio between balanced and unbalanced classes.

Each model was trained on two-thirds of the data and validated on the remaining third. After every iteration, accuracy, ROC-AUC, F1-score, and precision-recall metrics were computed. The results

from all three folds were then averaged to obtain a stable estimate of performance. Cross-validation reduced the influence of random data splits and increased the reliability of the evaluation, which was essential given the small dataset size.

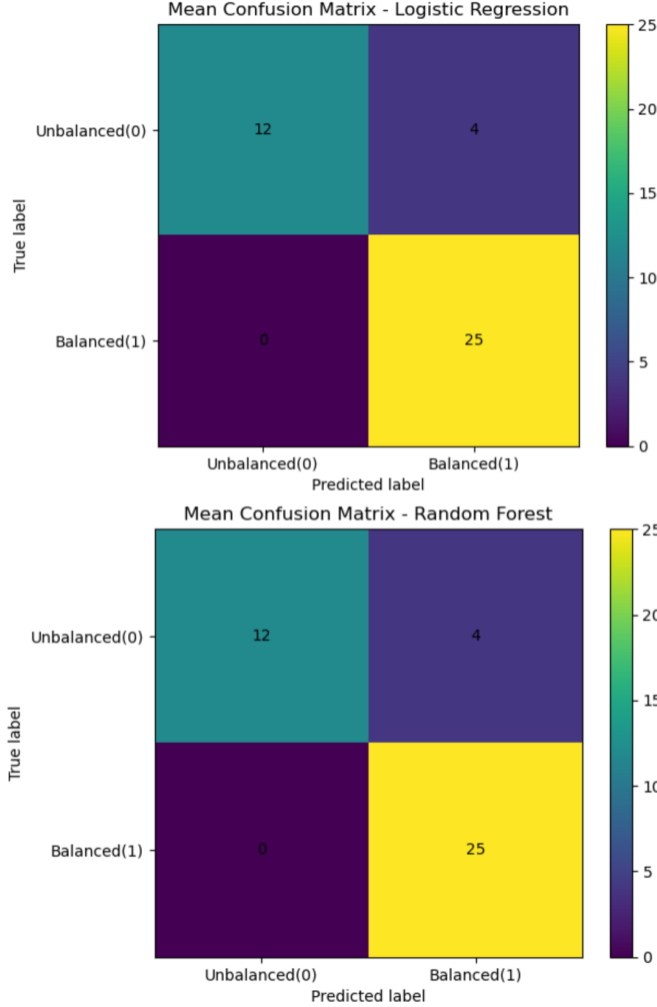


Figure 2: Mean confusion matrices for Logistic Regression and Random Forest models under three-fold cross-validation. Both classifiers correctly identified all balanced weapons and most unbalanced ones, demonstrating highly consistent prediction behavior.

Both Logistic Regression and Random Forest achieved strong and nearly identical results. Logistic regression reached a mean accuracy of approximately 0.90 ± 0.09 and a mean ROC-AUC of 0.96 ± 0.04 . Precision and recall were balanced between classes, with macro-F1 and weighted-F1 scores close to 0.9. The random forest model performed almost the same across all metrics. This similarity occurred because the dataset was relatively small and contained clear, strong feature signals. When the relationships between features and labels are largely linear and the data is noise-free, complex models like random forests often converge toward

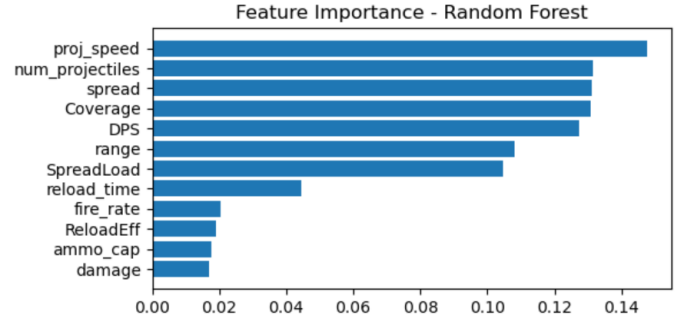


Figure 3: Feature importance ranking from the Random Forest model, showing the most influential weapon parameters for predicting perceived balance.

the same decision boundaries as simpler models such as logistic regression. The classification performance of both models is further illustrated in their mean confusion matrices, as shown in Figure 2.

The confusion matrix provided additional insight. All 25 balanced weapons were correctly identified, and twelve out of 16 unbalanced weapons were detected. The remaining unbalanced cases were misclassified as balanced, showing a minor bias toward predicting balance. In this context, such bias is acceptable because overlooking a well-balanced weapon would be more problematic for game design than misclassifying a slightly overpowered one.

To interpret the model’s reasoning, feature importance was analyzed using the random forest model (see Figure 3). The ranking revealed that projectile speed, number of projectiles, and spread had the highest influence on the predicted balance probability. Features such as coverage (product of projectile speed and range), damage per second (DPS), and range followed closely, reflecting their combined role in shaping offensive potential. Parameters related to weapon management, such as reload time, fire rate, and reload efficiency, contributed less but still added nuance to the prediction.

This ranking matches intuitive design expectations. Weapons with fast projectiles and multiple shots often feel powerful, but if their spread is too large or their range too long, they can appear unfair. Moderate parameter values across these top-ranked features tended to correspond to higher perceived balance scores.

After model validation, the random forest classifier was applied to predict the perceived balance of new weapon designs using the same nine quantitative parameters employed during model training: projectile speed, fire rate, damage, range, number of projectiles, spread, ammunition capacity, reload time, and critical chance. Derived metrics such as damage per second, reload efficiency, coverage, and spread load were generated in the same way as in the training process, ensuring consistency in feature representation.

For each new weapon configuration, the model outputs a balance probability score that represents how likely players would perceive the weapon as balanced. Probabilities above 0.7 indicate a well-balanced design, values between 0.5 and 0.7 suggest borderline cases that may need further tuning, and values below 0.5 imply that the weapon could be unbalanced. This predictive process demonstrates how the trained model can assist designers by providing immediate

```

1  # Create a new weapon
2  new_weapon = pd.DataFrame([
3      "proj_speed": 20,
4      "fire_rate": 5,
5      "damage": 15,
6      "range": 12,
7      "num_projectiles": 1,
8      "spread": 0,
9      "ammo_cap": 20,
10     "reload_time": 1.5,
11     "crit_chance": 0.05
12 ])
13
14 # Calculate derived features
15 new_weapon["DPS"] = new_weapon["damage"] * new_weapon["fire_rate"]
16 new_weapon["ReloadEff"] = new_weapon["ammo_cap"] / new_weapon["reload_time"]
17 new_weapon["Coverage"] = new_weapon["range"] * new_weapon["proj_speed"]
18 new_weapon["SpreadLoad"] = new_weapon["num_projectiles"] * new_weapon["spread"]
19
20 # Predict by random forest
21 prob = rf.predict_proba(new_weapon[FEATURES].values)[0, 1][0]
22 print(f"Predicted balance probability: {prob:.3f}")
23
24 if prob >= 0.7:
25     print("This weapon is likely balanced.")
26 elif prob >= 0.5:
27     print("Borderline balance (may need tuning).")
28 else:
29     print("Likely unbalanced (rework parameters).")
30
✓ [28] 13毫秒

```

Predicted balance probability: 0.794
This weapon is likely balanced.

Figure 4: Predicted balance probabilities for newly designed weapons tested using the trained Random Forest model. Probabilities above 0.7 indicate a well-balanced design, while those below 0.5 suggest potential imbalance.

feedback on the likely fairness of a weapon before new playtests are conducted.

In summary, the evaluation confirmed that the model successfully captured the relationship between weapon parameters and perceived balance. The strong agreement between the two classifiers demonstrates that the dataset contained consistent and interpretable patterns. The feature importance analysis provided actionable insights for design adjustment, and the prediction framework illustrated how these findings can be translated into a practical tool for supporting real-time weapon balancing decisions.

7 Discussion

The results show that player perceptions of weapon balance can be modeled effectively from a small set of quantitative parameters. Both logistic regression and random forest classifiers performed consistently well, indicating that the link between physical weapon attributes and subjective balance judgments is stable and predictable. This supports the broader view in game user research that measurable gameplay variables can represent player experience when combined with subjective feedback.

From a design perspective, the analysis identifies the features that most strongly shape perceptions of fairness. Projectile speed, number of projectiles, and spread were the most influential, defining

how responsive, powerful, or chaotic a weapon feels during play. Parameters such as damage and reload efficiency had less influence, suggesting that players judge balance more through moment-to-moment control than through long-term resource management. Recognizing this helps designers focus on the mechanical aspects that most affect player satisfaction.

The close agreement between the two models reflects both the simplicity of the dataset and the clarity of the underlying relationships. With only a few weapons and players, the patterns were strong enough that even a linear model could capture them. At the same time, this simplicity limits generalizability. A larger dataset with more diverse weapons would likely introduce non-linear effects that could be better modeled by a random forest or a neural approach.

7.1 Limitations

Small sample size. Only 13 participants and four weapons were included, which restricts statistical power and model diversity. A larger and more varied weapons and participant pool would capture a larger range of preferences and behaviors, providing a better basis for generalization.

External validation. The model has not yet been tested with new weapons or human participants beyond the initial study, so its real-world generalization remains unverified. Conducting follow-up tests with other data or additional player groups would be needed to confirm the validity of the model’s predictions.

7.2 Future Work

Future work should expand the dataset with additional players and weapon types to capture more complex interactions. Collecting more detailed in-game statistics, such as hit accuracy or kill-to-shot ratio, could enrich the model’s predictive features. Testing the trained model on new weapons and new participants would provide valuable validation of its generalizability. Finally, using regression instead of binary classification could allow for finer-grained predictions of perceived balance, revealing subtler patterns in how players experience fairness.

In summary, this project demonstrates that a small, well-structured dataset combining subjective and objective metrics can yield interpretable and meaningful predictions for weapon balance. It shows that data-driven modeling can complement traditional design intuition, offering a practical and analytical tool for improving fairness and engagement in game development.

8 Conclusion

This project explored how supervised learning can be used to predict perceived weapon balance in a 2D PvE shooting game. By combining subjective questionnaire responses with objective weapon parameters, we developed a model that links measurable design features to players’ experience of fairness and control. The process included constructing a weighted Perceived Balance Score, cleaning and engineering features from gameplay data, and training both logistic regression and random forest classifiers.

The evaluation showed that even with a small dataset, the models accurately predicted whether a weapon felt balanced, achieving

mean accuracies around 90 percent. Both classifiers performed similarly, suggesting that the relationship between weapon attributes and player perception was strong and largely linear. Feature importance analysis confirmed that projectile speed, number of projectiles, and spread were the dominant factors influencing balance judgments. These findings align closely with practical design intuition and provide quantitative evidence for key aspects of weapon tuning.

While the study was limited by sample size and the narrow scope of weapon types, it demonstrated the feasibility of using AI to support game-balancing decisions. The approach can help designers anticipate how players might experience new weapons before extensive playtesting. Future work could extend the dataset, include more diverse mechanics, and evaluate model generalization on new weapons and player groups.

In conclusion, this project shows that player-centered data and machine learning can complement traditional design intuition, offering a systematic and data-informed way to maintain fairness and engagement in game development.

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A Appendix

The first question of the questionnaire was whether the participant gave consent to take part in the study. Also, a block was added to add a Test number (which was important for the researchers). The participant got nine statements about each weapon, and they had to answer on a 7-scale whether they agreed with the statement or not. The statements addressed the four weapons in order: rifle, shotgun, SMG, and sniper rifle.

A.1 Consent Form

We are asking you to participate in a research study. We will describe this study to you and answer any of your questions. This study is being led by Binxu Jiang, Nick Jansen, Subienay Ganesh Ratnam, and Yating Qu, from Artificial Intelligence and Game and Media Technology, Faculty of Science, at Utrecht University. The purpose of this study is to investigate how different weapon types influence players’ perceived balance, enjoyment, and performance in a 2D player-versus-environment (PVE) shooting game. You will be asked to play a short 2D shooting game that includes **four distinct weapons**:

- Rifle
- Shotgun
- SMG
- Sniper Rifle

During gameplay, you can **switch between weapons at any time using the “Q” key**. Your main objective is to **defeat all enemies and clear four rooms within a 180-second time limit**. If you clear all rooms before the timer runs out, you complete the task successfully. If the timer expires before clearing all rooms, the task is considered failed.

After gameplay, you will be asked to complete a short **questionnaire** about your experience with each weapon, including your perceptions of their **balance, effectiveness, and enjoyment**. These responses will help determine whether each weapon is perceived as balanced.

The total duration of this test is approximately **5-7 minutes**.

We do not anticipate any risks from participating in this study.

By participating, you will gain firsthand experience with different weapon types and have the opportunity to compare and reflect on their **balance, performance, and play experience** within a short action scenario.

A.2 Questionnaire

- (1) When using this weapon, the level of challenge felt appropriate - neither too easy nor too hard.
- (2) This weapon feels effective but not overpowered for the current enemies.
- (3) Overall, this weapon feels well-balanced within the game environment.
- (4) I enjoyed using this weapon.
- (5) I felt a good sense of achievement when using this weapon
- (6) I would choose this weapon again in future sessions.
- (7) I am satisfied with the projectile speed (how fast projectiles travel).
- (8) I am satisfied with the weapon speed (how often I can fire or use it).
- (9) I am satisfied with the weapon damage (how much each projectile deals).
- (10) I am satisfied with the weapon range (how far projectiles can travel before disappearing).
- (11) I am satisfied with the ammunition and reload time (capacity and reloading pace).

A.3 Repository

The game repository is shared using GitFront. It is accessible using the following link:

<https://gitfront.io/r/NickJansen/y8TV3etABWow/TopDownShooter/>